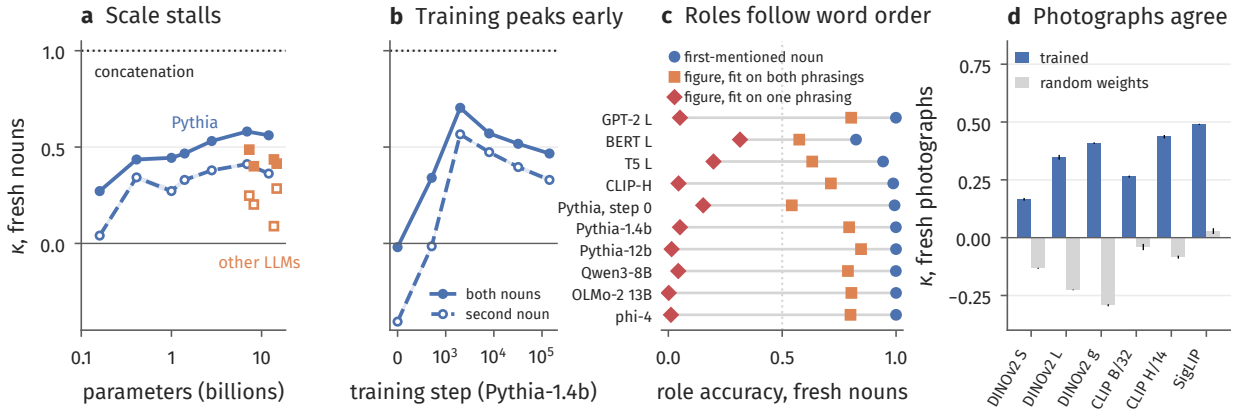

a Scale stalls
1.0
0.5
0.0
κ, fresh nouns
concatenation
Pythia
other LLMs
0.1
1
10
parameters (billions)
b Training peaks early
both nouns
second noun
0
10^3
10^4
10^5
training step (Pythia-1.4b)
c Roles follow word order
first-mentioned noun
figure, fit on both phrasings
figure, fit on one phrasing
GPT-2 L
BERT L
T5 L
CLIP-H
Pythia, step 0
Pythia-1.4b
Pythia-12b
Qwen3-8B
OLMo-2 13B
phi-4
0.0
0.5
1.0
role accuracy, fresh nouns
d Photographs agree
trained
random weights
0.75
0.50
0.25
0.00
−0.25
κ, fresh photographs
DINOv2 S
DINOv2 L
DINOv2 g
CLIP B/32
CLIP H/14
SigLIP